

QuantumGuard — drift detector benchmark report

Hybrid classical / quantum-inspired drift monitoring for a production fraud-detection model.

Held-out scenarios (calibration + feature-map selection): sudden_covariate_shift_v1, gradual_covariate_shift_v1

Evaluation scenarios (final numbers, disjoint — asserted): sudden_covariate_shift_v2, gradual_covariate_shift_v2, fraud_subset_shift_v1, no_drift_control_v1

Detection threshold: pre-drift mean + 3.0 sigma (held-out pre-drift windows only); persistence 2 consecutive windows.

Deployed quantum feature map (frozen before evaluation): angle

Generated by benchmark/export_pdf.py from results.csv, ablation_results.csv and INTERPRETATION.md.

Detection performance — population view

scenario	kind	detector	detected	delay	false_alarm_rate	separation
sudden_covariate_shift	sudden	classical_cds	True	0.0	0.0	140.73
sudden_covariate_shift	sudden	domain_classifier	True	0.0	0.0	30.98
sudden_covariate_shift	sudden	mmd_rbf	True	0.0	0.0	405.51
sudden_covariate_shift	sudden	quantum_angle	True	0.0	0.0	131.24
sudden_covariate_shift	sudden	quantum_reupload	True	0.0	0.0	87.3
sudden_covariate_shift	sudden	quantum_zz	True	0.0	0.0	126.07
gradual_covariate_shift	gradual	classical_cds	True	2.0	0.0	109.06
gradual_covariate_shift	gradual	domain_classifier	True	4.0	0.0	21.22
gradual_covariate_shift	gradual	mmd_rbf	True	3.0	0.0	323.3
gradual_covariate_shift	gradual	quantum_angle	True	2.0	0.0	328.78
gradual_covariate_shift	gradual	quantum_reupload	True	3.0	0.0	153.85
gradual_covariate_shift	gradual	quantum_zz	True	3.0	0.0	173.14
fraud_subset_shift_v1	fraud_subset	classical_cds	True	1.0	0.0	4.87
fraud_subset_shift_v1	fraud_subset	domain_classifier	True	16.0	0.0	2.45
fraud_subset_shift_v1	fraud_subset	mmd_rbf	True	29.0	0.0	1.39
fraud_subset_shift_v1	fraud_subset	quantum_angle	True	26.0	0.0	2.57
fraud_subset_shift_v1	fraud_subset	quantum_reupload	False	—	0.0	1.98
fraud_subset_shift_v1	fraud_subset	quantum_zz	True	26.0	0.0	2.21
no_drift_control_v1	none	classical_cds	False	—	0.0	—
no_drift_control_v1	none	domain_classifier	False	—	0.0	—
no_drift_control_v1	none	mmd_rbf	False	—	0.0	—
no_drift_control_v1	none	quantum_angle	False	—	0.0	—
no_drift_control_v1	none	quantum_reupload	False	—	0.0	—
no_drift_control_v1	none	quantum_zz	False	—	0.0	—

Detection performance — fraud-subset view

Windows below the minimum fraud-subset size are excluded (NaN-skipped).

scenario	kind	detector	detected	delay	false_alarm_rate	separation
sudden_covariate_shift	sudden	classical_cds	True	0.0	0.0	4.67
sudden_covariate_shift	sudden	domain_classifier	True	0.0	0.0	12.5
sudden_covariate_shift	sudden	mmd_rbf	True	0.0	0.0	75.45
sudden_covariate_shift	sudden	quantum_angle	True	0.0	0.0	23.42
sudden_covariate_shift	sudden	quantum_reupload	True	0.0	0.0	20.39
sudden_covariate_shift	sudden	quantum_zz	True	0.0	0.0	20.21
gradual_covariate_shift	gradual	classical_cds	True	19.0	0.0	2.9
gradual_covariate_shift	gradual	domain_classifier	True	10.0	0.0	10.58
gradual_covariate_shift	gradual	mmd_rbf	True	10.0	0.0	60.55
gradual_covariate_shift	gradual	quantum_angle	True	1.0	0.0	13.55
gradual_covariate_shift	gradual	quantum_reupload	True	5.0	0.0	11.14
gradual_covariate_shift	gradual	quantum_zz	True	1.0	0.0	12.34
fraud_subset_shift_v	fraud_subset	classical_cds	True	1.0	0.0	4.77
fraud_subset_shift_v	fraud_subset	domain_classifier	True	1.0	0.0	11.69
fraud_subset_shift_v	fraud_subset	mmd_rbf	True	1.0	0.0	60.04
fraud_subset_shift_v	fraud_subset	quantum_angle	True	1.0	0.0	24.02
fraud_subset_shift_v	fraud_subset	quantum_reupload	True	1.0	0.0	22.03
fraud_subset_shift_v	fraud_subset	quantum_zz	True	1.0	0.0	19.13
no_drift_control_v1	none	classical_cds	False	—	0.0	—
no_drift_control_v1	none	domain_classifier	False	—	0.0	—
no_drift_control_v1	none	mmd_rbf	False	—	0.0102	—
no_drift_control_v1	none	quantum_angle	False	—	0.0	—
no_drift_control_v1	none	quantum_reupload	False	—	0.0	—
no_drift_control_v1	none	quantum_zz	False	—	0.0	—

Feature-map ablation (held-out scenarios only)

angle = no-entanglement control; selection frozen before evaluation.

feature_map	scenario	subset	detected	delay	false_alarm_rat	separation	mean_wall_clock_ms
angle	sudden_covariate_shi	population	True	0	0.0	36.45	13.7
angle	sudden_covariate_shi	fraud	True	0	0.0	18.66	11.6
zz	sudden_covariate_shi	population	True	0	0.0	38.22	96.0
zz	sudden_covariate_shi	fraud	True	0	0.0	18.21	62.2
reupload	sudden_covariate_shi	population	True	0	0.0	29.75	34.6
reupload	sudden_covariate_shi	fraud	True	0	0.0	16.71	26.2
angle	gradual_covariate_shi	population	True	1	0.0	104.51	15.3
angle	gradual_covariate_shi	fraud	True	4	0.0	16.43	11.3
zz	gradual_covariate_shi	population	True	1	0.0	103.68	110.0
zz	gradual_covariate_shi	fraud	True	4	0.0	13.26	68.8
reupload	gradual_covariate_shi	population	True	1	0.0	68.49	39.7
reupload	gradual_covariate_shi	fraud	True	4	0.0	14.26	29.4

Compute cost — mean wall-clock ms per window

detector	mean_wall_clock_ms
quantum_angle	15.6
domain_classifier	27.4
classical_cds	29.8
mmd_rbf	36.1
quantum_reupload	44.7
quantum_zz	114.9

Interpretation

Written after reviewing results.csv and ablation_results.csv; regenerating REPORT.md re-embeds this file verbatim. Numbers below are from the frozen evaluation run (thresholds and feature map selected on held-out scenarios only).

Does the quantum-inspired kernel add anything?

On population-level drift: no. Every detector catches the sudden shift at delay 0 and the gradual shift within 2–4 windows, all with zero false alarms on the control. The classical per-feature CDS is as fast as any kernel method, and on the fraud-subset-shift scenario viewed at population level it is dramatically better (delay 1 vs 16–29): per-feature quantile-bin statistics notice a small shifted subpopulation that whole-distribution kernel methods dilute away. Data re-uploading missed that scenario outright at population level.

On small-sample fraud-subset monitoring: yes, clearly. On the gradual scenario's fraud-subset windows ($n \approx 40$ per window), the quantum kernels detect at delay 1 (angle, ZZ) versus 10 for MMD-RBF and the domain classifier and 19 for the classical CDS. This is the one regime where the fidelity kernel earns its place: with few samples per window, the smooth angle-embedding kernel accumulates evidence of a mixture shift faster than a median-heuristic RBF or a cross-validated classifier can. Combined with the delay-1 detection on the fraud-only-shift scenario (tied with all others), the fraud-subset QDS is the differentiated signal that justifies the hybrid MHS design.

Ablation: entanglement does not help

The angle embedding — the no-entanglement control whose kernel factorizes into per-feature $\cos^2$ terms — ties ZZ/IQP on every held-out delay, edges it on separation (44.0 vs 43.3), and runs $6.5\times$ faster (13ms vs 84ms per window). Data re-uploading is strictly worse (lowest separation, one population-level miss). The honest conclusion: the benefit comes from having a smooth, bounded, well-conditioned kernel on PCA-reduced data, not from entanglement — the selected map is efficiently classically simulable, so this is "quantum-inspired" in the precise sense: the quantum formalism guided a good kernel choice, and no quantum advantage is claimed.

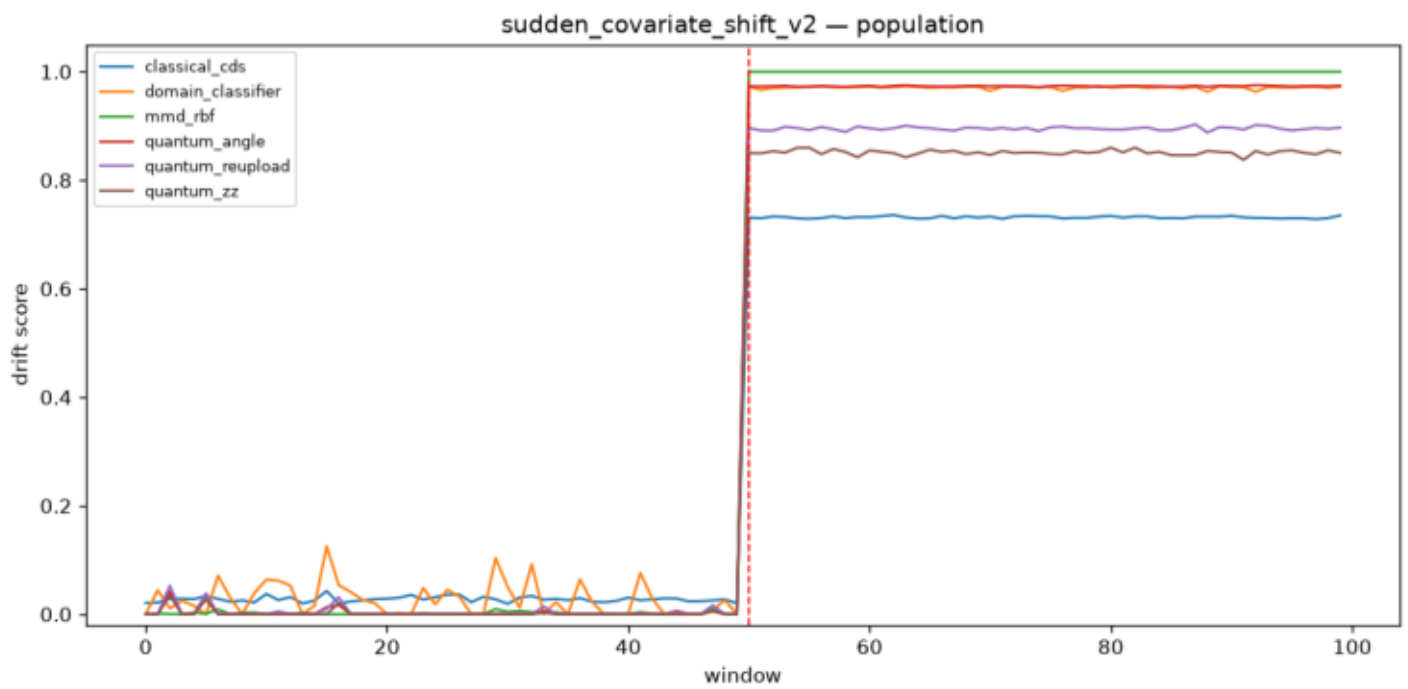
Cost

The selected angle map is the *cheapest* multivariate detector in the suite (~13–20ms per window, vs ~30ms MMD, ~25–35ms domain classifier, ~100–150ms ZZ), so the fraud-subset QDS signal comes at no compute premium.

Recommendation

Deploy `angle` for QDS (frozen in config before evaluation). Keep the classical per-feature CDS as the primary population signal, and treat the fraud-subset quantum kernel score as the early-warning channel for gradual drift in the fraud class — the scenario class that population monitoring is structurally blind to.

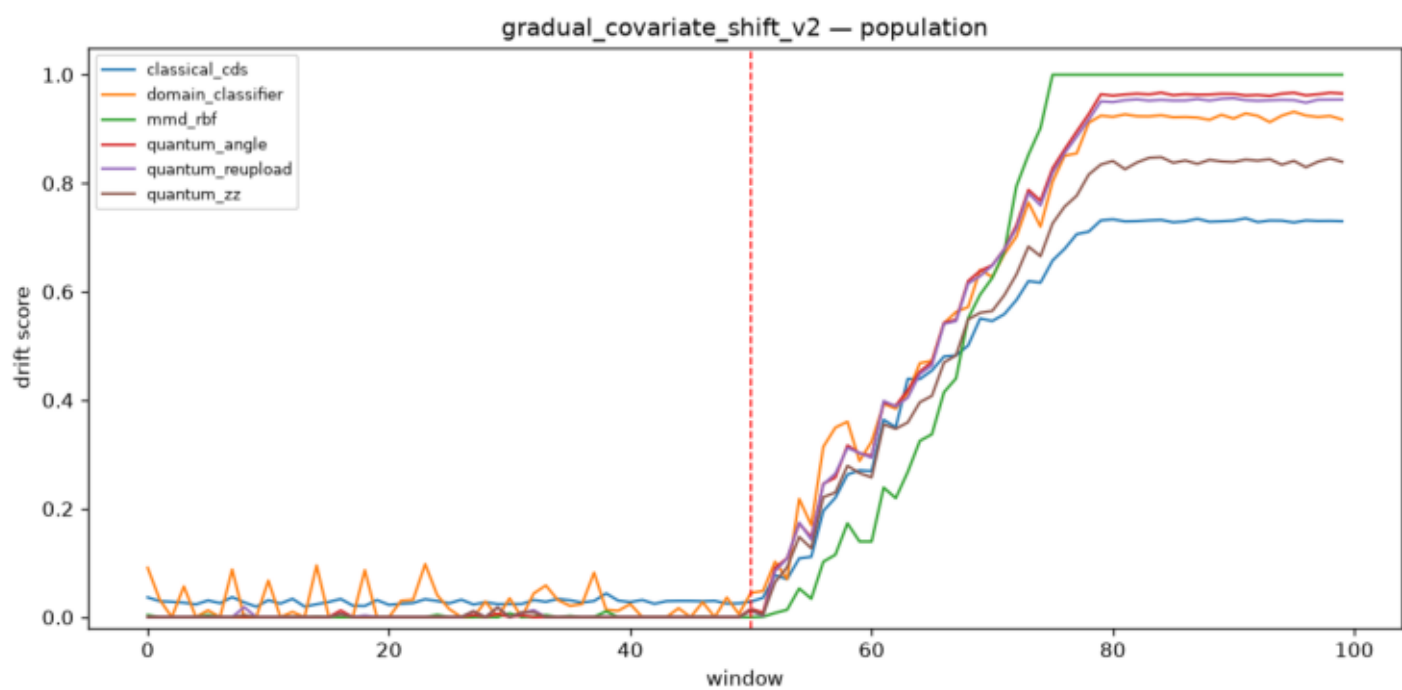
sudden_covariate_shift_v2 — population



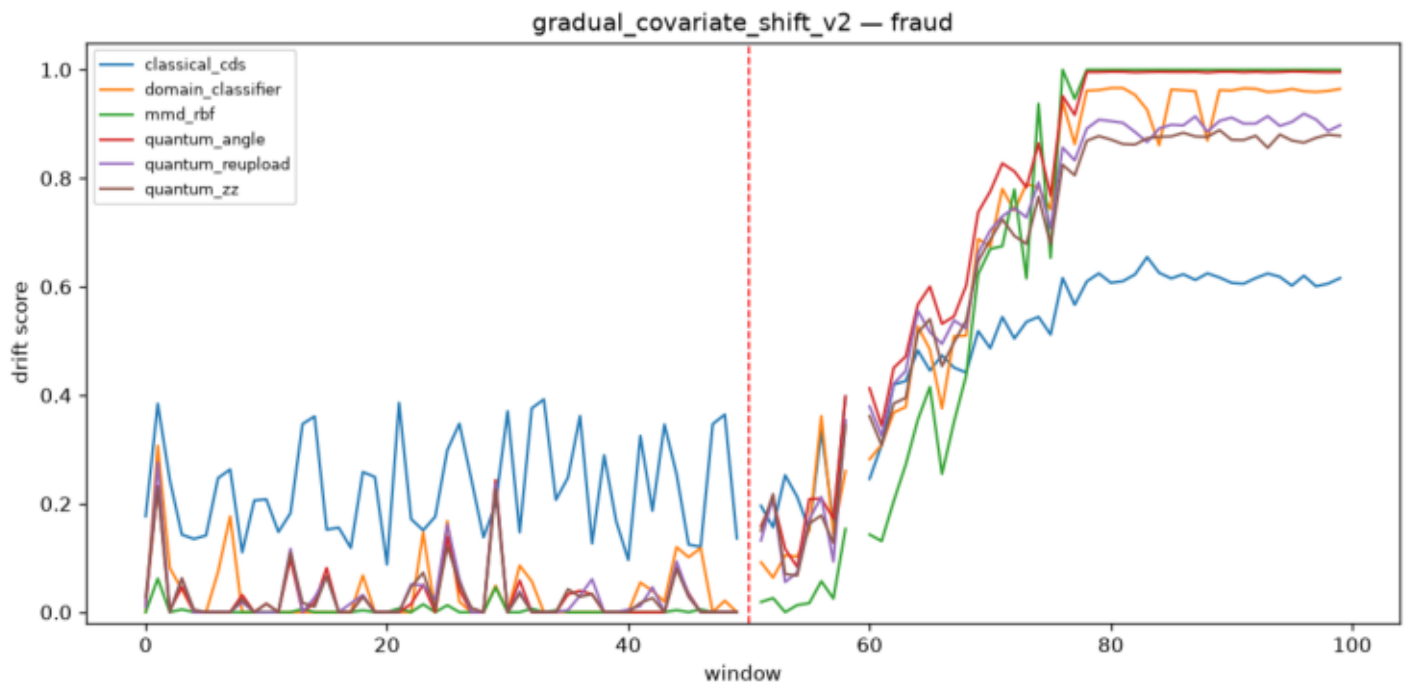
sudden_covariate_shift_v2 — fraud



gradual_covariate_shift_v2 — population



gradual_covariate_shift_v2 — fraud



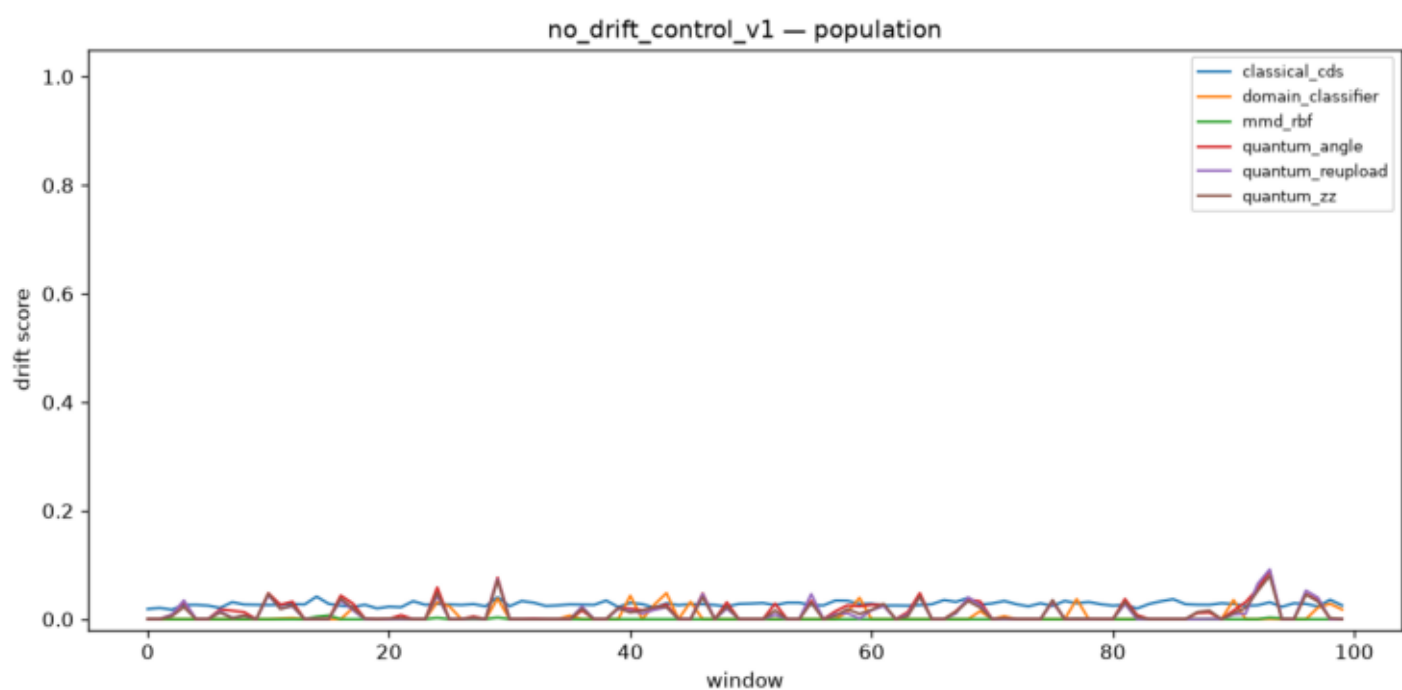
fraud_subset_shift_v1 — population



fraud_subset_shift_v1 — fraud



no_drift_control_v1 — population



no_drift_control_v1 — fraud

